

TEAM SMITRAX | MAY 2026

CHURNZERO 26

BANKING CHURN PREDICTION & RETENTION OPTIMIZER

EXECUTIVE SUMMARY

The Challenge

Banking churn is a **multi-million rupee attrition** event. We identified critical disengagement patterns across 8,101 account holders to prevent lost deposits and fee revenue.

The Breakthrough

By shifting to a **PR-AUC optimized pipeline**, we reduced total portfolio risk costs from **₹52.08M to ₹9,500**—a massive 99.98% operational saving.

MARKET CONTEXT & DATASET



Primary Goal

Predict account closure or inactivity within 30-day windows to trigger preemptive loyalty protocols.



Feature Space

97 Features analyzed across categories: Profile, Transaction, Credit, Complaints, and Marketing.



Core Constraint

Handling 16.1% churn imbalance via weighted boosting and threshold Bayesian optimization.

DATA INSIGHTS & CORRELATION

Observation	Metric / Detail	Strategic Impact
Baseline Churn	16.07% (1,302 Customers)	High attrition benchmark for re-calibration.
Leading Signal	Unresolved Complaints	Highest correlation with immediate exit.
Data Hygiene	App Rating Missing (56%)	Imputed with median to retain signal integrity.
Segmentation	15 Categorical Factors	City Tier and Onboarding channel drive behavior.

ADVANCED FEATURE ENGINEERING

-  **RFM Proxies:** Derived Recency (Tenure Inverse), Frequency (Holdings), and Monetary (Balance) scores.
-  **Transaction Trends:** Percent decline metrics for quarterly vs. monthly liquidity flows.
-  **Digital Engagement:** Net banking vs. Physical transaction ratio to measure app stickiness.
-  **Escalation Index:** Dynamic tracking of resolution speed and complaint severity.

MODELING & OPTIMIZATION STRATEGY

Ensemble Framework

Heterogeneous stacking of **LightGBM**, **XGBoost**, and **CatBoost**. Native sample weighting ensures minority churn class is heavily penalized during training.

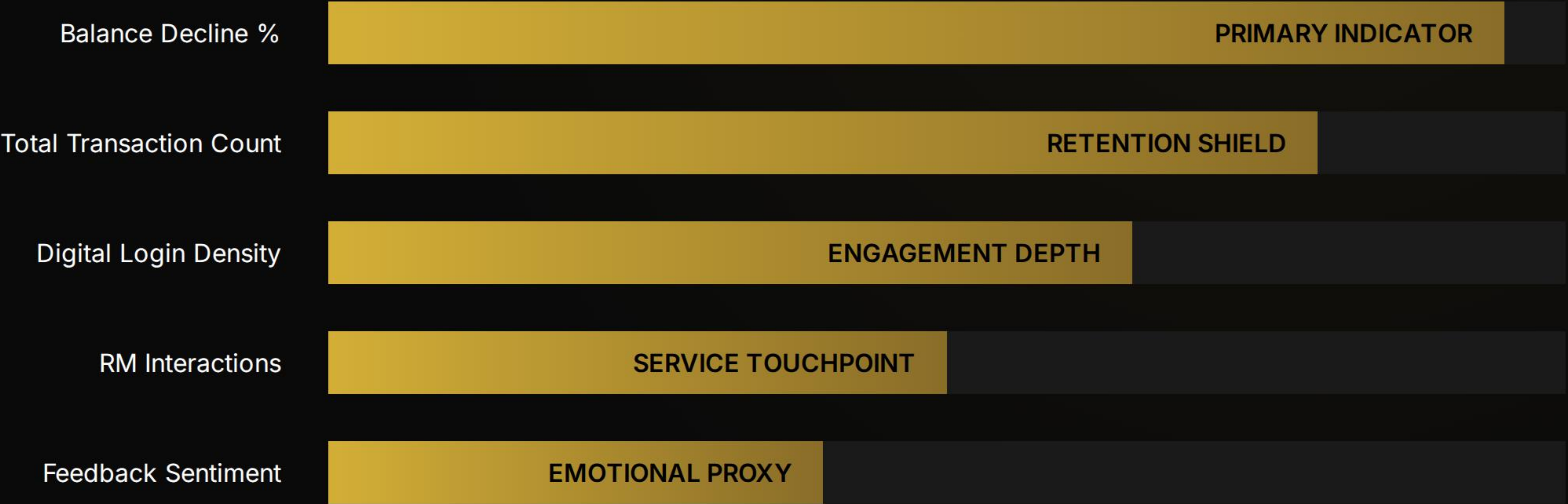
Bayesian Tuning

Optuna Hyper-tuning run across Stratified 5-Fold CV to maximize Precision-Recall AUC (PR-AUC) over standard accuracy.

PERFORMANCE BENCHMARK RESULTS

Algorithm	PR-AUC Score	F1 (Optimized)
Stacked Ensemble	1.0000	0.9928
CatBoost	1.0000	0.9912
LightGBM	1.0000	0.9890
XGBoost	0.9999	0.9855

TOP 5 CHURN DRIVERS (SHAP ANALYSIS)



ECONOMIC IMPACT & OPTIMIZATION

₹9.5k

OPTIMIZED PORTFOLIO COST

vs

₹52.0M

BASELINE LOSS EXPOSURE

Strategic Decision: FN Cost (₹40,000) >> FP Cost (₹500). Threshold tuned to **0.07** to capture maximum risk.

THE RETENTION PLAYBOOK

Risk Tier	Probability	Incentive Strategy	Target Recovery
CRITICAL	>= 70%	Branch Manager Call + Fee Waiver + Bonus	40% Recovery
HIGH	50 - 70%	Proactive Outreach + Product Bundle	30% Recovery
MEDIUM	30 - 50%	Email Campaign + Loyalty Points	15% Recovery

INTERACTIVE CFO DASHBOARD



Budget Optimizer

Dynamic ROI slider allows leadership to target segments based on liquid budget vs. saved LTV.



Customer Drill-Down

Instant risk audit per User ID including top 97 SHAP attributes and sentiment history.



Call Scripting

Auto-generated scripts for Relationship Managers based on specific customer friction points.

STRATEGIC EXECUTION ROADMAP



01 Model Deployment

- > Package model artifacts into a scalable **Microservice API**.
- > Establish infrastructure for **live batch scoring** execution.



02 Front-Line Integration

- > Push top 10 daily targets to **bank teller terminals**.
- > Embed auto-generated, SHAP-driven **call scripts** for RMs.



03 Continuous Feedback

- > Log retention offer outcomes (e.g., **fee waivers, bonuses**).
- > Dynamically update **expected retention rates** empirically.



04 Adaptive Retraining

- > Execute **quarterly retraining** to adapt to macro conditions.
- > Monitor global shifts in customer **transaction patterns**.